

VALENCE AGREEMENT AS A TOP-DOWN CUE TO ANAPHORA RESOLUTION

Kalle Glauch
Ruhr-Universität Bochum

RUB

Hypothesis

Anaphora resolution involves both bottom-up (e.g., Ariel, 1991; Gundel et al., 1993) and top-down cues. We propose that valence agreement between the anaphora and potential antecedents is a key top-down cue: the more their valences match, the more likely the anaphora will target that referent. This hypothesis stems from two psychological processes: inferring character traits from available information (e.g., Newman et al., 2014; Sripada, 2010), and interpreting behaviors as consistent with these traits (e.g., Leising et al., 2014; Sadler & Woody, 2003).

Study 1

Research Question: Are ambiguous agentive anaphoras more likely to target potential antecedents whose valence matches the valence of the subsequent action?

Method: 2x2 within-subject design: NP2 Valence (positive vs. negative) and Action Valence (positive vs. negative). Participants read sentences like “The NP1 of the NP2 who ACTION...” and selected pronoun target between NP1 and NP2. The valence rating were taken from the sentiment dictionaries of Warriner et al. (2013) and Mohammad (2018).

Items: 40 experimental items in 4 conditions distributed using a Latin-Square Design. NP1 consists of relational nouns.

POS – POS: The neighbor of the gentleman who supported a friend...
POS – NEG: The neighbor of the gentleman who humiliated a friend...
NEG – POS: The neighbor of the jailer who supported a friend...
NEG – NEG: The neighbor of the jailer who humiliated a friend...

Participants: Participants (n = 61; Age: $M = 41$, $SD = 16$; Gender: female = 37 females, male = 19, non-binary = 4) were recruited via Prolific.

Data Exclusion: Participants (n = 8) with $\geq 25\%$ of response times ≤ 1000 ms were excluded. Participants (n = 13) with 100% responses for same NP were excluded. Additional 86 individual observations with a response time of ≤ 1000 ms were removed.

Results: A generalized linear mixed-effects model (GLMM) was fitted to the data using the R package *lme4* (Douglas Bates et al., 2015).

`glmer(interpretation ~ NP2_Valence * Action_Valence + (1 | participant) + (1 | item), data = data, family = binomial(link = "logit"))`

The model reveals a significant main-effect of NP2-Valence ($p = 0.001^{***}$) and its interaction with Action-Valence ($p = 0.001^{***}$), but no significant main-effect of Action-Valence ($p = 0.698$).

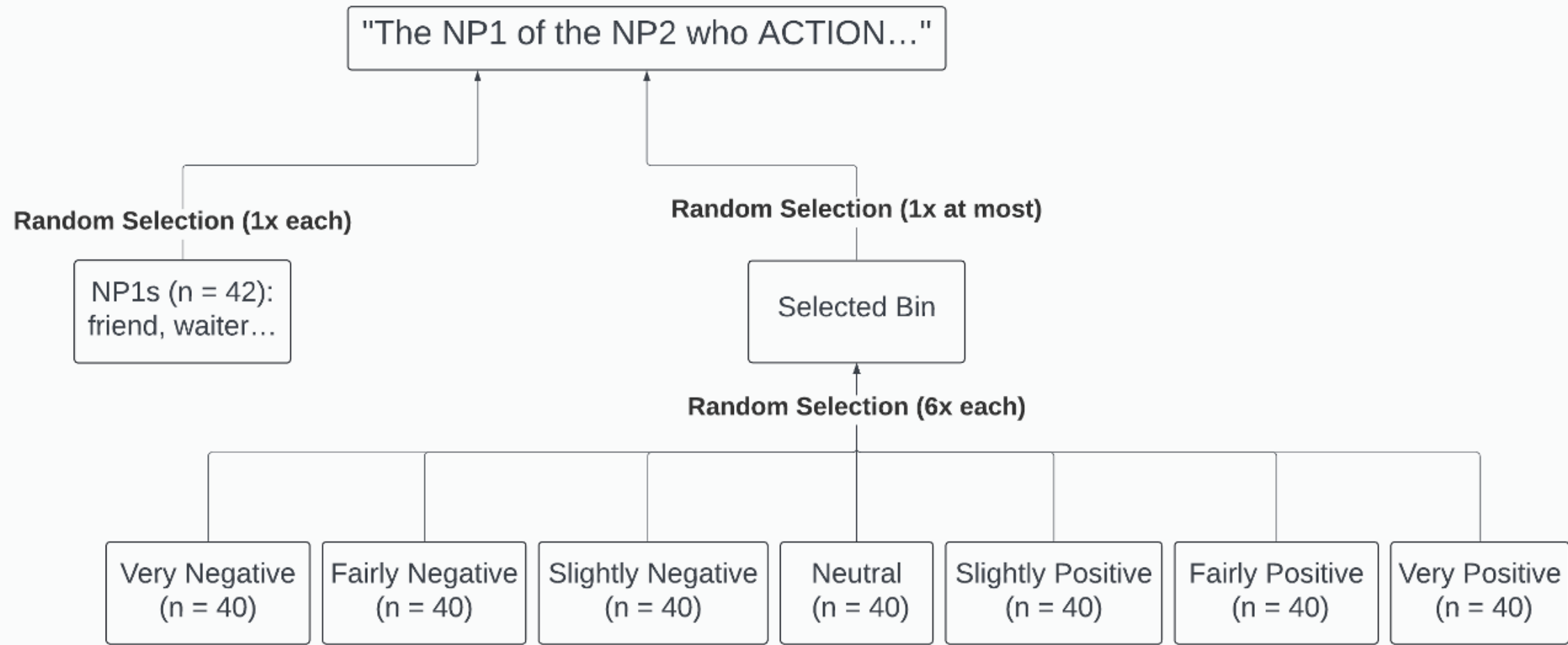
Discussion:

- Valence agreement significantly influences pronoun resolution.
- The effect is stronger for negative referent-action pairs than for positive ones, indicates polarity-specific differences in valence association strength.

Study 2

Research Question: How does valence agreement influence anaphora resolution across varying ranges of valence differences?

Method: Same task as in Study 1. Combination of classical list design and random word selection. Only *Action Valence* (very positive vs. very negative) was manipulated in a factorial manner. NP1 and NP2 were randomly drawn from a predefined set of words for each trial, automatically executed for each experimental run. This approach allowed for investigating fine-grained valence differences without needing many lists and prevented idiosyncratic NP1-NP2-ACTION associations from skewing results.



Items: 42 experimental items were distributed across lists. Below are examples of randomly generated sentences during the experiment:

The tenant of the gentleman (-2.94) who fairly shared the credit...
The secretary of the astronaut (-1.55) who protected the endangered wildlife...
The driver of the student (-0.02) who mocked a friend's ambition...
The client of the nutcase (1.52) who bullied a vulnerable child...
The plumber of the gambler (2.41) who deliberately broke the car window...
The teacher of the villain (4.37) who squandered the investment money...

Participants: Participants (n = 59; Age: $M = 41$, $SD = 16$; Gender: female = 33, male = 23, non-binary = 3) were recruited via Prolific

Data Exclusion: Participants (n = 8) with $\geq 25\%$ of response times ≤ 1300 ms were excluded. Participants (n = 11) with 100% responses for same NP were excluded. Additional 13 individual observations with a response time of ≤ 1300 ms were removed.

Results: A generalized additive mixed-effect model (GAMM) was fitted to the data using the R package *mgcv* (Wood & Wood, 2015).

`gam(interpretation ~ s(valence_difference, by = valence_action, k = 10) + valence_action + s(participant, bs = "re"), family = binomial(link = "logit"), data = data)`

The model shows a significant non-linear interaction between negative actions and valence difference ($p = 0.005^{**}$, $edf = 2.9$). In contrast, the interaction with positive actions is not significant ($p = 0.17$).

Discussion:

- Valence agreement significantly influences anaphora resolution for negative actions but not for positive actions, indicating polarity-specific differences in valence association.
- For negative actions, the effect of valence agreement is mostly linear: greater valence difference between potential antecedents increases the likelihood that the pronoun targets the agreeing referent.

Observed Proportion of NP2 Readings by NP2 Valence and Action Valence with 95% Two-Sided CI



References

- Ariel, M. (1991). The function of accessibility in a theory of grammar. *Journal of Pragmatics*, 16(5), 443–463. [https://doi.org/10.1016/0378-2166\(91\)90136-L](https://doi.org/10.1016/0378-2166(91)90136-L)
- Douglas Bates, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48.
- Gundel, J. K., Hedberg, N., & Zacharski, R. (1993). Cognitive status and the form of referring expressions in discourse. *Language*, 69, 274–307. <https://doi.org/10.2307/416535>
- Leising, D., Gallrein, A.-M. B., & Dufner, M. (2014). Judging the behavior of people we know: Objective assessment, confirmation of preexisting views, or both? *Personality and Social Psychology Bulletin*, 40(2), 153–163.
- Mohammad, S. M. (2018). Obtaining reliable human ratings of valence, arousal, and dominance for 20,000 english words. *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*.
- Newman, G. E., Bloom, P., & Knobe, J. (2014). Value judgments and the true self. *Personality and Social Psychology Bulletin*, 40(2), 203–216.
- Sadler, P., & Woody, E. (2003). Is who you are who you're talking to? interpersonal style and complementarity in mixed-sex interactions. *Journal of Personality and Social Psychology*, 84(1), 80–96. <https://doi.org/10.1037/0022-3514.84.1.80>
- Sripada, C. S. (2010). The deep self model and asymmetries in folk judgments about intentional action. *Philosophical Studies*, 151, 159–176. <https://doi.org/10.1007/s11098-009-9423-5>
- Warriner, A. B., Kuperman, V., & Brysbaert, M. (2013). Norms of valence, arousal, and dominance for 13,915 english lemmas. *Behavior Research Methods*, 45(4), 1191–1207. <https://doi.org/10.3758/s13428-012-0314-x>
- Wood, S., & Wood, M. S. (2015). Package 'mgcv'. *R package version*, 1(29), 729.

GAMM Predicted Proportion of NP2 Readings by Valence Difference and Action Valence

